

OOPs QUICK REVISION CHEAT SHEET (JAVA)

◆ 1. CLASS & OBJECT

- Class → Blueprint
- Object → Instance of class (new keyword)

ClassName obj = new ClassName();

- Default values:
 - int → 0
 - boolean → false
 - reference → null
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◆ 2. METHODS

- Syntax:

returnType methodName(params)

- Method Signature = Name + Parameters (NOT return type)
- Entry point:

public static void main(String[] args)

◆ 3. CONSTRUCTORS

- Same name as class
- No return type
- Used for initialization

Types:

- Default
- Parameterized

 Important:

- Constructor can be overloaded
 - Constructor NOT inherited
 - If not defined → default constructor added
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◆ 4. THIS & SUPER

- this → current object
- super → parent class

this.x = x;

super.show();

◆ 5. INHERITANCE

- Keyword: extends

Types:

- Single
- Multilevel
- Hierarchical

 Multiple inheritance (with classes) not supported

◆ 6. POLYMORPHISM

✓ Compile-time (Static)

- Method Overloading

✓ Runtime (Dynamic)

- Method Overriding

 Important:

- Methods → polymorphic
- Variables → NOT polymorphic

- Static methods → NOT polymorphic
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◆ 7. METHOD OVERRIDING RULES

- Same name + same parameters
 - Runtime binding
 - Cannot override:
 - final
 - private
 - static
 - Access level → same or less restrictive
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◆ 8. ENCAPSULATION

- Data hiding using:

private + getters/setters

```
int getX() { return x; }
```

```
void setX(int x) { this.x = x; }
```

🔥 Trap:

x = x; ❌

this.x = x; ✓

◆ 9. ABSTRACTION

- Hides implementation

Achieved by:

- Abstract class
 - Interface
-

◆ 10. ABSTRACT CLASS

- Cannot create object
 - Can have:
 - Abstract methods
 - Normal methods
 - Constructor
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◆ 11. INTERFACE

- Methods → public abstract (default)
- Variables → public static final

✓ Supports multiple inheritance

Java 8+:

- default methods
- static methods

Java 9+:

- private methods
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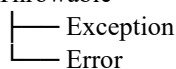
◆ 12. EXCEPTION HANDLING

Keywords:

- try, catch, finally, throw, throws

Hierarchy:

Throwable



✓ Checked (compile-time)

- IOException

✓ Unchecked (runtime)

- NullPointerException

🔥 Important:

- finally always runs (except System.exit)

♦ 13. GARBAGE COLLECTION

- Removes unused objects
- Runs automatically

System.gc(); // request only

♦ 14. ACCESS MODIFIERS

Modifier Same Class Same Package Subclass Other

private	✓	✗	✗	✗
default	✓	✓	✗	✗
protected	✓	✓	✓	✗
public	✓	✓	✓	✓

♦ 15. OBJECT CLASS (VERY IMPORTANT)

- Parent of all classes

Important methods:

- toString() → Class@hashcode
- equals() → comparison
- hashCode()

♦ 16. MOST IMPORTANT TRAPS ⚠️

- 👉 Static method + null reference → Works
- 👉 Variable access → depends on reference
- 👉 Method call → depends on object
- 👉 Wrong catch type → Runtime crash
- 👉 Missing this → setter fails
- 👉 final method → cannot override
- 👉 Interface conflict → must override

🔥 FINAL 10-SECOND REVISION

- Overloading → compile-time
- Overriding → runtime
- Variables ✗ polymorphism
- Static ✗ overriding
- finally → always runs
- private → no inheritance/override
- interface → multiple inheritance
- abstract ≠ object creation
- this vs super (VERY COMMON)
- getters/setters (watch `t